

# Executive Summary

HOYA BIT Trust Layer  
PoC5 Evidence → 8-12 Pilot → 3-6 Production  
SLAauditbudgettenantdashboard  
KPI run API usageseatenterprise integration  
live/cache/fixture API EvidenceLogartifact

## 

RAG Crypto AI BI Dashboard TrustForge  
Claim → Evidence → Trust → Report

### HURRICANESOFT

TrustForge Hermes

Evidence-native Decision Intelligence Agent

|  | HurricaneSoft |  |            |
|--|---------------|--|------------|
|  |               |  | 2026-08-01 |
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TrustForge AI AI Agent

## 

TrustForge Hermes evidence-native decision intelligence agent  
evidence contract Claims deterministic trust

reasoning Amazon Bedrock TrustedBrief decision record TrustForge Trust Layer

TrustForge Trust Score

- 
- claim-to-source
- APIartifact telemetry
- providermemorypolicycalibrator backend

RAG

|  | ingestionclaim extractiondeduplication evidence grouping | Analysis JourneyFinal Report                |
|--|----------------------------------------------------------|---------------------------------------------|
|  | uncertaintylimits could-flip                             | Evidence ListTrust breakdown                |
|  | run_idExecution Logmodel version manifest                | Execution LogSHA-256 manifest               |
|  | learning event outcome gate                              | Training contractcandidate approval receipt |

TrustForge LLM prompt

2.1

| Ingestion Layer        |             | pricesHOYA BITnewsocialregulatoryon-chain whale tradesCoinGeckoCMCDefiLlamaEtherscan    |
|------------------------|-------------|-----------------------------------------------------------------------------------------|
| Reasoning Layer        | uncertainty | pipelineTrust Kernelcorroborationevidence grouperfreshnessdirection resolution          |
| Agent / Delivery Layer | API         | Hermes orchestratorBedrockreport deliveryFinal ReportEvidence ListExecution LogSnapshot |

2.2

Frontend React/Vite Hermes Workspace Backend Python APIanalysis flow formal run runtime API run\_id

| Research Workspace  | AnalyzeCompareHistorySource StatusWhale ActivityEvidence Trail facts/inferences ladder report download |
|---------------------|--------------------------------------------------------------------------------------------------------|
| Operations Console  | Training StatusUpgrade QueueModule TelemetryBudgetCarbonHealth                                         |
| Governance Console  | AdminAGOSpolicymemoryprompt versionapprovalrollbackaudit record                                        |
| Integration Surface | API responsesnapshotartifactmanifestFinal ReportEvidence ListExecution LogSource/Config                |

2.3 AWS

EC2 t3.micro + nginx nginx HTTPS frontendPython API 127.0.0.1:8080Amazon Bedrock LLM S3 IAM Instance RoleSSM Session Manager CloudWatch App Runner

| コンポーネント                | 機能                                                           | 依存関係                           |
|------------------------|--------------------------------------------------------------|--------------------------------|
| EC2 + nginx            | React/Vite frontend, Python API reverse proxy                | 証明書/TLS, 127.0.0.1 API binding |
| Amazon Bedrock         | claim extraction, stance classification, report assembly, 推論 | 信頼スコア (Trust Score)            |
| S3                     | 保存された artifact, manifest ファイル                                | IAM 権限                         |
| IAM / SSM / CloudWatch | SSH key, ログ監視                                                | セキュリティ                         |

## アーキテクチャ

### 3.1 Data Ingestion

Ingestion は様々な source から fetched\_at, content\_reference, asset, data\_mode, freshness を記録し、データを蓄積する。

- HOYA BIT から OHLCV データを delayed outcome ground truth として取得
- RSS, API, X, Reddit から様々なデータを取得
- Etherscan, Whale Alert, Arkham, CoinMarketCap から様々なデータを取得
- 様々なデータを蓄積する
- cache, safe fetch, connector reliability, freshness, data integrity を SHA-256 manifest で検証

### 3.2 Trust Kernel

Trust Kernel は deterministic trust reasoning を用いて LLM の出力を評価し、confidence と uncertainty を算出する。

Trust Kernel は Amazon Bedrock を利用して推論を行う。

### 3.3 Formal Run

- formal\_run\_coordinator は budget reservation, idempotency, lease を管理
- analysis flow は claim extraction, stance, deduplication, corroboration, Trust Kernel, uncertainty を実行

- `run` `run_id` `Final Report``Evidence List``Execution Log``Snapshot``Source/Config` `API response`
- `timeout``degraded``abstain` `fail-closed`

### 3.4 Agent RuntimeMemory Policy

Hermes orchestrator AgentCore adapter/runtime tool registry memory OS policy compiler/executor Agent RuntimeMemory plane `run``artifact``evidence``learning event` `replay history``policy plane` `source` `analysis` `report` `evaluation` `improvement`

AgentCore TrustForge domain logic TrustForge `contract``guards` `audit`

### 3.5 Upgrade Control

Improvement `upgrade control` `canary` `activation lock` `deployment evidence` `rollback record` `refresh_sources` `measure_quality` `replay_history` `diagnose_improvement` `candidate` `candidate benchmark` `activation`

| Candidate | production                                          | metricsartifact hashtraining contract           |
|-----------|-----------------------------------------------------|-------------------------------------------------|
| Review    | qualitysecuritycostcompatibility adversarial review | reviewer attestationdecision record             |
| Activate  | canary activation lock                              | approvaldeployment evidence receipt             |
| Rollback  |                                                     | rollback decision recordhealth checkaudit trail |

### 3.6

- Module Telemetry
- Budget Guard `token``Bedrock` `Agent` `USD`
- Carbon `input/output token` `Trust Kernel` `LLM`

- Audit hermes audit execution events artifact registry authenticated ledger

## Bedrock

### 4.1

Bedrock foundation model prompt

- deterministic feature prompt
- delayed outcome confidence benchmark
- ModelHub governance SageMaker execution backend artifact
- foundation model Amazon Bedrock TrustForge task-specific calibrator foundation model

### 4.2 Learning Event

learning event Claims Trust Score raw confidence run\_id Execution Log

outcome T+1 T+7 T+14 1714

### 4.3 backend

1. Run learning event immutable evidence execution metadata
2. T+1/T+7/T+14 outcome label training row label leakage
3. Isotonic Regression candidate ECE Brier holdout
4. SageMaker backend AWS S3 artifact ModelHub
5. candidate review canary activation production

training trigger CLI scheduler run activation enhancement

### 4.4 ModelHub SageMaker

| Tool                         | Category                   | Capabilities                                                                                        |
|------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------|
| <a href="#">ModelHub</a>     | Governance surface         | model version, training request, metrics, artifact provenance, candidate status, approval lifecycle |
| <a href="#">SageMaker AI</a> | Training execution backend | Amazon S3, Amazon SageMaker Training Job, Amazon S3 artifact, Amazon SageMaker                      |
| S3                           | Artifact storage           | training dataset, calibrator artifact, manifest, Amazon S3                                          |

## 5.0 Introduction

### 5.1 Data Sources

| Source      | Category                     | Capabilities                                     |
|-------------|------------------------------|--------------------------------------------------|
| OHLCV       | HOYA BIT, prices, CoinGecko  | Historical T+ outcome, real-time                 |
| News        | RSS, API, news connector     | Real-time news, sentiment analysis               |
| Social      | X, Reddit, social connector  | Real-time social media, sentiment analysis       |
| Blockchain  | Etherscan, Solscan, on-chain | Real-time blockchain data, sentiment analysis    |
| Regulatory  | Taiwan regulatory            | Real-time regulatory updates, sentiment analysis |
| Market Data | CMC, CoinGecko, DefiLlama    | Real-time market data, sentiment analysis        |

### 5.2 Evidence Contract

The [Claim](#) object contains source, fetched\_at, content\_reference, asset, data\_mode, and related\_claim. The [Evidence List](#) object contains a list of evidence objects, each containing source, fetched\_at, content\_reference, asset, data\_mode, and related\_claim.

### 5.3 Data Sources

| 項目                         | 内容                                    | 備考 |
|----------------------------|---------------------------------------|----|
| Final Report               | Trust Scoreconfidencelimitscould-flip |    |
| Evidence List              | claim_id                              |    |
| Execution Log              |                                       |    |
| Source / Config + Snapshot | manifest                              |    |

- API keyAWS credentialsGitHubIAM Instance RoleSSM secret boundary
- tool registry policy guards scope
- Fail-closeddegraded abstain
- run\_idmodel versionartifact hashapproval receiptdeployment evidence rollback record
- candidate adversarial reviewsecret production gate

TrustForge ISO/IEC 42001 ISO 42001



Overview

| Phase                               | Key Components                                         | Key Metrics                                 |
|-------------------------------------|--------------------------------------------------------|---------------------------------------------|
| Phase 1: Research Trust Layer       | Trust Layer API, report, evidence, execution, snapshot | Trust Layer API adoption, review turnaround |
| Phase 2: Governed Decision Services | API, peer metrics, asset context                       | API adoption, review turnaround             |
| Phase 3: Continuous Calibration     | outcome label, candidate, ModelHub, SageMaker, canary  | ECE, Brier, candidate, rollback             |
| Phase 4: Enterprise Control Plane   | IAM, audit, budget, carbon, telemetry, policy          |                                             |

Key Features

- Evidence contract, claim-to-source lineage, prompt
- Trust Kernel, backend
- Bounded autonomous agent
- Calibration, replay, artifact provenance, rollback
- 

Architecture

| Component    | Details                                                                     |
|--------------|-----------------------------------------------------------------------------|
| data_mode    | data_mode, fixture, cache, live                                             |
| Trust        | Trust, Evidence                                                             |
| Final Report | Final Report, Evidence List, Execution Log, Source/Config, manifest, run_id |
| training row | training row, outcome label, candidate, metrics, artifact hash              |
| degraded     | degraded, abstain, fail-closed                                              |

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- TrustForge 環境構築
- 各 provider (AgentCore, DynamoDB backend) のインストール
- 各 provider の run 環境の構築 (scheduler, activation)
- 各 provider のテスト

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[illegible]

# AI Agent

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|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|

| 項目                            | 説明                      |
|-------------------------------|-------------------------|
| Evidence-native               | 証拠が生成された環境で実行される        |
| Deterministic trust reasoning | 信頼の推論が決定論的である           |
| Training backend              | 学習のバックエンドは artifact である |
| Candidate                     | 候補の生成に使用される             |
| Audit trail                   | 監査の履歴を記録する              |
| Fail-closed                   | 失敗した場合は閉鎖状態になる          |

TrustForge Hermes HurricaneSoft